

Statistical Mechanics

60 marks

End Semester Exam

3 hrs

1. (a) Using statistical mechanics, derive the mean occupation number $\langle n \rangle$ for bosons (Bose-Einstein statistics) and fermions (Fermi-Dirac statistics). 3
(b) Prove that in the high-temperature limit both distributions reduce to the Maxwell-Boltzmann form. [8 marks] 2
2. (a) Define and discuss the Gibbs entropy. 1.5
(b) Show that it reduces to the Boltzmann entropy formula for a microcanonical ensemble and that it is maximised by the Boltzmann distribution in the canonical ensemble. [4 marks] 1.5
3. In class, we derived the partition function for an ideal gas. Now consider a more realistic system in which (i) each particle excludes a finite volume, and (ii) each particle experiences an average attractive interaction due to all other particles.
(a) Construct the canonical (NVT) partition function for this system by incorporating these effects as perturbations to the ideal gas. 3
(b) From this partition function, derive expressions for the Helmholtz free energy and the pressure. [10 marks]
(c) Hence, obtain the resulting equation of state for the system. 2
4. (a) For a single quantum harmonic oscillator, write down the canonical partition function and derive an expression for the mean energy. Show explicitly that, in the high-temperature limit, the result reduces to the classical equipartition prediction. 1.5
(b) In the Einstein model of solids, a crystalline solid is modeled as N independent quantum harmonic oscillators of identical frequency. Using this model, derive an expression for the heat capacity C_v . Analyze its behavior in the high-temperature and low-temperature limits. [8 marks] 2

5. A reaction coordinate (or collective variable) is often used to describe a chemical or physical process in a system under canonical (NVT) conditions.

(a) Starting from the canonical partition function, explain how the potential of mean force (PMF) can be computed as a function of the reaction coordinate.

(b) Discuss the physical significance of the PMF and how its features relate to stable states and transition states.

(c) Describe how reaction rates can be estimated from the PMF, clearly stating any assumptions involved. [6 marks]

6. Consider the two-dimensional Ising model on a square lattice.

(a) Briefly describe the model, including its Hamiltonian and key physical features. 2

(b) Outline a Monte Carlo scheme (e.g., Metropolis algorithm) to sample equilibrium configurations of the system. Provide pseudocode for the algorithm. 3

(c) Using the sampled configurations, explain how you would compute the mean energy, heat capacity, and average magnetization. 1.5

(d) Write down the canonical partition function for a 50×50 times 50×50 Ising lattice. 1.5

(e) Express the mean energy, heat capacity, and magnetization in terms of derivatives of the partition function. [10 marks] 2

7. Consider a system in the grand canonical (μVT) ensemble, described by the grand partition function. In class, we derived the expression for energy fluctuations in the canonical ensemble and established their relationship with the heat capacity. Analogously, you need to derive an expression for particle-number fluctuations in the grand canonical ensemble.

(a) Starting from the grand partition function, derive an expression for the mean particle number $\langle N \rangle$. 4

(b) Express the fluctuation in particle number ($\langle N^2 \rangle - \langle N \rangle^2$) in terms of derivatives of the grand partition function. [14 marks] 5

(c) Prove that $\langle N^2 \rangle - \langle N \rangle^2 = k_B T (\partial \langle N \rangle / \partial \mu)_T$ 2.5